Distributions

How the pooled observed intensities differ from a normal distribution.

AT A GLANCE

Where	Data → Distribution
Data levels	Protein, phosphosite, phosphoprotein
Draws	One Q-Q plot: every observed value of the annotated $\log_2$ matrix against the normal distribution.
Matrix	The annotated $\log_2$ matrix, always. There is no **Data Source** selector, so filtering and imputation do not change what is drawn; normalization does.
Sampling	Above 100,000 observed values a seeded random sample of 100,000 is plotted, and a warning says so.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The Q-Q plot compares ordered observed intensities with theoretical normal quantiles. An approximately straight pattern is consistent with a normal shape; curvature shows how the shapes differ. This is a visual diagnostic, not a pass/fail test of the analysis pipeline. An upward bend at the high end means the largest observations exceed the fitted normal reference. Check the scale, sample distributions and upstream missing-value handling before assigning a cause.

All sample columns are pooled. This mixes features with different abundances and samples

from different conditions. It does not assess the residuals for each feature or the within-pair differences in a paired test, which are the relevant distributions for those parametric tests. Pearson correlation can be calculated without normal intensities; its inferential assumptions are a separate question.

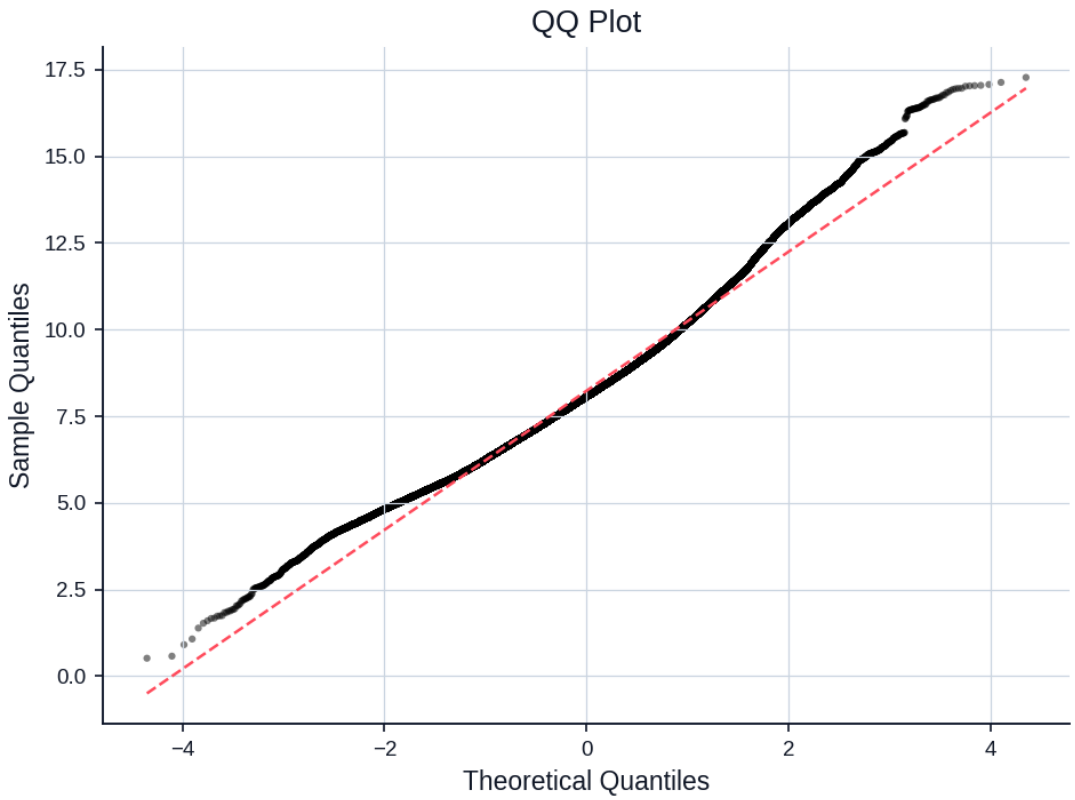


Figure 1.1. The page's own Q-Q plot on the example data, at protein level: sample quantiles of the $\log_2$ matrix, running from about 0 to 17.5, against the theoretical quantiles of the normal distribution, with the fit line through the plotted points. The cloud lies above the line at the left end and above it again at the right while the middle of the range sits on it, and there is no step or shelf anywhere along it.

Reading it

- One point, one observed value. Its horizontal position is the quantile a normal distribution would put it at, its vertical position the value itself.
- The red line: a least-squares straight line through the plotted points. It is the line the points would lie on if they were normal, fitted to them rather than assumed.
- The ends. Points above the line at the right and below it at the left are a distribution with heavier tails than normal; the reverse is lighter tails. A bend at one end only is a skew in that direction.
- A step or a shelf near the lower end is usually a floor in the data — a detection limit, a filled value, an imputation already carried in the upload — rather than a property of the biology.

The page carries a second panel, **QQ Norm Reference**, which is a fixed figure with its own

explanatory text. It is part of the application, is the same whatever is loaded, and is there to be compared against the plot above.

> **EASY TO GET WRONG**
>
> A curve in the Q-Q plot is not automatically a problem. A proteomics matrix pools tens of thousands of features whose abundances span several orders of magnitude, and the mixture of them is not expected to be exactly normal; mild departure at both ends is the normal case. What is worth acting on is a departure in the middle of the range, a step, or two visibly separate straight segments — the last usually means the matrix contains two populations, for example samples normalized differently, or a level mixed with values that are ratios rather than abundances.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phospho or phosphoprotein. Only the levels actually uploaded are offered, and the choice is remembered for the session.	Protein	page
Random seed	Fixes which values are drawn when the matrix is larger than the sampling limit, so the same figure is reproduced exactly. Below the limit it changes nothing.	187	page
Refresh	Reloads the summary behind the page, including the sampling warning.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Marker style	Size, colour, shape and opacity of the points.	small black dots, half opaque	appearance
Reference line	The fit line: whether it is drawn at all, and its colour, dash pattern and width.	shown, red, solid	appearance

WORTH KNOWING

The appearance panel is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**), and is described in Part XI. This plot has no legend, no conditions and no sample labels, so the panel offers none of those cards here; the title, axis, font and background cards work as they do everywhere else.

3 How it is computed

HOW IT IS COMPUTED

The sample columns of the annotated $\log_2$ matrix are coerced to numbers and flattened into one vector, and every non-finite entry — NA and any infinity — is dropped. Nothing else is removed: a zero and a negative value are kept, which matters because a $\log_2$ intensity is negative wherever the linear intensity is below 1, and a phosphoprotein matrix collapsed with robust_site_median holds ratios centred on zero.
If more than 100,000 values remain, 100,000 of them are drawn without replacement, using a generator seeded with **Random seed**.
The remaining n values are sorted ascending and become the sample quantiles. The theoretical quantile of the i-th of them is

$$t_i = \Phi^{-1}\left(\frac{i - 0.5}{n}\right), \qquad i = 1, \dots, n,$$

with Φ^{-1} the inverse standard normal distribution function. Each point is $(t_i, x_{(i)})$.

THE FIT LINE

The line is the ordinary least-squares fit of the sample quantiles on the theoretical quantiles — the straight line through *all* the plotted points, not a line through two chosen quantiles — drawn from the smallest theoretical quantile to the largest. Its slope is therefore an estimate of the standard deviation of the values and its intercept an estimate of their mean, and the ends of the cloud pull on it: a heavy lower tail tilts the line, which is why a departure is read as a shape rather than as a distance from a fixed reference.

Edge cases

- Conditions and replicates play no part. All samples are pooled, so a condition with one replicate is drawn exactly like any other.

- Missing values are not plotted and not replaced; they simply do not enter the vector.
- Fewer than two usable values gives one point and no fit line. No usable values at all is an error rather than an empty figure.
- Where the values are all equal the line is horizontal, which is the correct picture of a matrix with no spread.
- Because the page reads the annotated matrix, running filtering or imputation does not change this figure. Regenerating the annotation with a different **Normalization** does.

4 The three shapes

Beside the figure the page shows a reference panel: three quantile–quantile plots of data whose distributions are known, so that the shape on screen can be compared with a shape whose meaning is settled. They are the same three here.

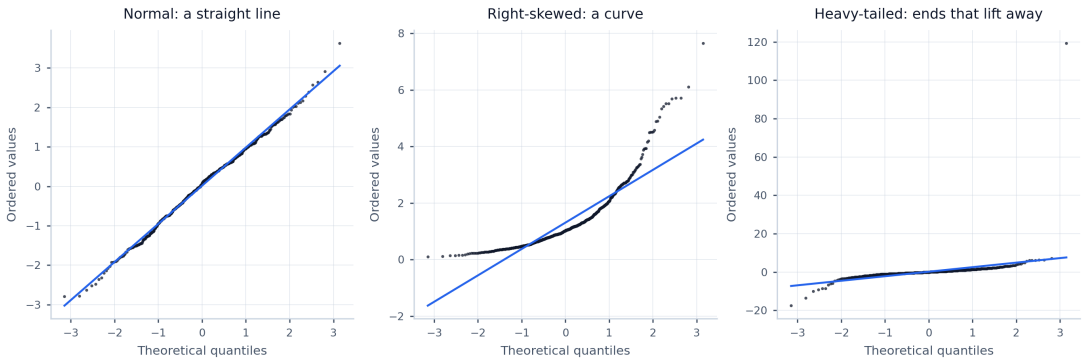


Figure 1.2. The three shapes a quantile–quantile plot takes. Left: values drawn from a normal distribution. Middle: values with a long right tail. Right: values from a distribution with heavy tails at both ends.

An approximately straight pattern is consistent with a normal shape for the pooled observations. It does not prove normality or validate every downstream test. Log transformation often reduces right skew, but neither transformation nor normalization guarantees a normal distribution.

Asymmetric curvature can indicate skewness. A long upper tail is one possible pattern. Verify the declared scale, but do not infer that it is wrong or that all p-values are optimistic from this plot alone. A mixture of valid feature distributions can be curved.

Ends that lift away from the line while the middle stays on it means heavy tails: more extreme values, in both directions, than a normal distribution would produce. In a proteomics matrix this is ordinary in moderation — the dynamic range is real, and a handful of very high and very low values is the expected consequence. It matters when it is not moderate: extreme values pull means and standard deviations, so a fold change or a t-statistic computed over a

feature whose values sit in such a tail rests on one or two measurements. The opposite, ends that bend *towards* the line, means the values are truncated — a quantitation floor, a filter applied upstream, or a report that writes a fixed value in place of a low measurement.

> **WORTH KNOWING**
>
> Read the middle of the plot first and the ends second. A departure in the central mass of the points says something about the whole matrix; a departure confined to the last few points on either side describes a handful of features and is usually not worth acting on.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = qqnorm_plot(data, meta)
```

R

```
fig <- qqnorm_plot(data, meta)
```

One call and one argument beyond the frames: `seed` defaults to 187, the same value the page's **Random seed** starts at, so the sampling that kicks in above 100,000 observed values draws the same subset in a notebook as on screen. What the packages do not carry is the rest of the page — there is no reference panel of the three shapes beside the figure, and no `distribution_summary` counts under it, so the comparison the page sets up has to be made against this chapter instead.

6 Exporting

This is an in-app diagnostic. It intentionally has no plot download or notebook-download button. The notebook function above remains available for an independently written analysis; it is not a code export from this page.

7 Further reading

NIST: probability plots explains comparison with theoretical quantiles. A pooled intensity plot is not a substitute for checking the assumptions of a particular statistical model.

Behind this page

Function	What it produces
qqnorm_plot	The Q-Q plot in both packages, including the 100,000-value sampling and its seed
qqnorm_plot_data	The points and the fit line as numbers
distribution_qqnorm_plot	The figure this page shows
distribution_summary	The counts behind the page: observed values, missing values, rows with and without a missing value, and the sampling warning